

EDFA host computer

1. Data transmission (hexadecimal transmission, baud rate 9600, 1 stop bit, no parity bit, 8 data bits)

HEAD1	HEAD2	LEN	ADDR	DATA	SUM
-------	-------	-----	------	------	-----

HEAD: 0xEF 0xEF

LEN: Data length (number of DATA bits + 2)

ADDR: Data address bit

DATA: Data bit (DATA is empty when querying)

SUM: Sum check $SUM = HEAD1 + HEAD2 + LEN + ADDR + DATA$

2. Data reception (hexadecimal reception, baud rate 9600, 1 stop bit, no parity bit, 8 data bits)

HEAD1	HEAD2	LEN	ADDR	DATA	SUM
-------	-------	-----	------	------	-----

HEAD: 0xED 0xFA

LEN: Data length (number of DATA bits + 2)

ADDR: Data address bit

DATA: Data bits

SUM: Sum check $SUM = HEAD1 + HEAD2 + LEN + ADDR + DATA$

3. Query device status (data address bit 0x00)

Sending Commands:

0xEF	0xEF	0x02	0x00	0xE0
------	------	------	------	------

Return command:

0xED	0xFA	0x0E	0x00	DATA(1~12)	SUM
------	------	------	------	------------	-----

Current Read = $DATA1 * 256 + DATA2$ (mA)

Input Power Read = $(DATA5 * 256 + DATA6) / 100 - 70$ (dBm)

Output Power Read = $(DATA7 * 256 + DATA8) / 100 - 70$ (dBm)

Example:

Sending Commands: EF EF 02 00 E0

Return commands: ED FA 0E 00 03 E8 00 00 17 70 23 28 00 00 00 00 B2

Current Read = $0x03 * 256 + 0xE8 = 1000$ (mA)

Input Power Read = $(0x17 * 256 + 0x70) / 100 - 70 = -10.00$ (dBm)

Output Power Read = $(0x23 * 256 + 0x28) / 100 - 70 = 20.00$ (dBm)

4. Query target power (data address bit 0x03)

Sending Commands:

0xEF	0xEF	0x02	0x03	0xE3
------	------	------	------	------

Return commands:

0xED	0xFA	0x04	0x03	DATA(1~2)	SUM
------	------	------	------	-----------	-----

Target Power = $(\text{DATA1} \times 256 + \text{DATA2}) / 100 - 70$ (dBm)

Example:

Sending Commands: EF EF 02 03 E3

Return commands: ED FA 04 03 23 39

Target Power = $(0x23 \times 256 + 0x28) / 100 - 70 = 20.00$ (dBm)

6. Set target power (data address bit 0x04)

Sending Commands:

0xEF	0xEF	0x04	0x04	DATA(1~2)	SUM
------	------	------	------	-----------	-----

Return commands:

0xED	0xFA	0x04	0x03	DATA(1~2)	SUM
------	------	------	------	-----------	-----

Target Power = $(\text{DATA1} \times 256 + \text{DATA2}) / 100 - 70$ (dBm)

Example:

Sending Commands: EF EF 04 04 23 27 30

Return commands: ED FA 04 03 23 27 38

Target Power = $(0x23 \times 256 + 0x27) / 100 - 70 = 19.99$ (dBm)

7. Query working mode (data address bit 0x05)

Sending Commands:

0xEF	0xEF	0x02	0x05	0xE5
------	------	------	------	------

Return commands:

0xED	0xFA	0x03	0x05	DATA	SUM
------	------	------	------	------	-----

Work Mode = DATA (0x00 for APC, 0x01 for ACC, 0x02 for AGC)

Example:

Sending Commands: EF EF 02 05 E5

Return commands: ED FA 03 05 00 EF

Work Mode = 0x00 (APC)

Return commands: ED FA 03 05 01 F0

8. Set the working mode (data address bit 0x06)

Sending Commands:

0xEF	0xEF	0x03	0x06	DATA	SUM
------	------	------	------	------	-----

Return commands:

0xED	0xFA	0x03	0x05	DATA	SUM
------	------	------	------	------	-----

Work Mode=DATA (0x00 for APC, 0x01 for ACC, 0x02 for AGC)

Example:

Sending Commands: EF EF 03 06 00 E7

Return commands: ED FA 03 05 00 EF

Work Mode=0x00 (APC)

9. Query target current (data address bit 0x07)

Sending Commands:

0xEF	0xEF	0x02	0x07	0xE7
------	------	------	------	------

Return commands:

0xED	0xFA	0x06	0x07	DATA(1~4)	SUM
------	------	------	------	-----------	-----

Current=DATA1*256+DATA2 (mA)

Example:

Sending Commands: EF EF 02 07 E7

Return commands: ED FA 06 07 01 F4 00 00 E9

Current=0x01*256+0xF4=500 (mA)

10. Query the maximum current limit (data address bit 0x09)

Sending Commands:

0xEF	0xEF	0x02	0x09	0xE9
------	------	------	------	------

Return commands:

0xED	0xFA	0x06	0x09	DATA(1~4)	SUM
------	------	------	------	-----------	-----

Current Limitation=DATA1*256+DATA2 (mA)

Example:

Sending Commands: EF EF 02 09 E9

Return commands: ED FA 06 09 21 34 00 00 4B

Current=0x21*256+0x34=8500 (mA)

11. Set the target current (data address bit 0x0C)

Sending Commands:

0xEF	0xEF	0x04	0x0C	DATA(1~2)	SUM
------	------	------	------	-----------	-----

Return commands:

0xED	0xFA	0x06	0x07	DATA(1~4)	SUM
------	------	------	------	-----------	-----

Current=DATA1*256+DATA2 (mA)

Example:

Sending Commands: EF EF 04 0C 01 F3 E2

Return commands: ED FA 06 07 01 F3 00 00 E8

Current=0x01*256+0xF3=499 (mA)

If the set value exceeds the maximum current limit, the current target current is returned.

12. Query the die temperature (data address bit 0x0B)

Sending Commands:

0xEF	0xEF	0x02	0x0B	0xEB
------	------	------	------	------

Return commands:

0xED	0xFA	0x06	0x0B	DATA(1~4)	SUM
------	------	------	------	-----------	-----

LD Temp=(DATA1*256+DATA2)/100(°C)

Example:

Sending Commands: EF EF 02 0B EB

Return commands: ED FA 06 0B 09 C4 00 00 C5

LD Temp=(0x09*256+0xC4)/100=25.00(°C)

13. Query target gain (data address bit 0x10)

Sending Commands:

0xEF	0xEF	0x02	0x10	0xF0
------	------	------	------	------

Return commands:

0xED	0xFA	0x04	0x10	DATA(1~2)	SUM
------	------	------	------	-----------	-----

Gain=(DATA1*256+DATA2)/100 (dB)

Example:

Sending Commands: EF EF 02 10 F0

Return commands: ED FA 04 10 07 D0 D2

Gain =(0x07*256+0xD0)/100 = 20.00 (dB)

14. Set target gain (data address bit 0x11)

Sending Commands:

0xEF	0xEF	0x04	0x11	DATA(1~2)	SUM
------	------	------	------	-----------	-----

Return commands:

0xED	0xFA	0x04	0x10	DATA(1~2)	SUM
------	------	------	------	-----------	-----

$$\text{Gain} = (\text{DATA1} * 256 + \text{DATA2}) / 100 \text{ (dB)}$$

Example:

Sending Commands: EF EF 04 11 07 D1 CB

Return commands: ED FA 04 10 07 D1 D3

$$\text{Gain} = (0x07 * 256 + 0xD1) / 100 = 20.01 \text{ (dB)}$$

13. Soft activation query (address bit 0x25)

Sending Commands:

0xEF	0xEF	0x02	0x25	0x05
------	------	------	------	------

Return commands:

0xED	0xFA	0x03	0x25	DATA	SUM
------	------	------	------	------	-----

SoftActive=DATA 1-Active 0-Shutdown

Example:

Sending Commands: EF EF 02 25 05

Return commands: ED FA 03 25 01 10 active ED FA 03 25 00 0F Shutdown

14. Soft Activation Setting (Address 0x26)

Sending Commands:

0xEF	0xEF	0x03	0x26	DATA	SUM
------	------	------	------	------	-----

Return commands:

0xED	0xFA	0x03	0x25	DATA	SUM
------	------	------	------	------	-----

SoftActive=DATA 1-Active 0-Shutdown

Example:

Setting the laser activation

Sending Commands: EF EF 03 26 01 08

Return commands: ED FA 03 25 01 10

Set laser off

Sending Commands: EF EF 03 26 00 07

Return commands: ED FA 03 25 00 0F

The activation command is valid only when the laser is in normal state and the key is turned on. .